1 Theoretical Framework

1.1 Signal model and scope

We consider compression of sampled power spectral density (PSD) frames. At each discrete time instant t, a sensing node computes a PSD estimate over a monitored frequency band $\Omega = [\omega_{\min}, \omega_{\max}]$ and must transmit a compact representation to a remote server under a bitrate constraint.

Let $\{\omega_n\}_{n=1}^N$ denote a uniform frequency grid over Ω, and define the sampled PSD frame

$$\mathbf{s}_t \triangleq \begin{bmatrix} s_{t,1} & \cdots & s_{t,N} \end{bmatrix}^\top \in \mathbb{R}_+^N, \qquad s_{t,n} \triangleq S_t(\omega_n). \tag{1}$$

The cloud reconstructs

$$\hat{\mathbf{s}}_t \triangleq \begin{bmatrix} \hat{s}_{t,1} & \cdots & \hat{s}_{t,N} \end{bmatrix}^\top \in \mathbb{R}_+^N. \tag{2}$$

The framework is composed of three parts:

1. a deterministic preprocessing transform with explicit side information,
2. a learned discrete latent codec with an operational rate model, and
3. a distortion design consisting of a core PSD fidelity term and an optional task regularizer.

1.2 Deterministic preprocessing and explicit side information

To separate deterministic simplification from learned compression, we define the preprocessing map

$$\mathbf{u}_t = \mathcal{T}_{\phi_p}(\mathbf{s}_t; \boldsymbol{\tau}_t) = \mathcal{S}_{\boldsymbol{\tau}_t}(\Psi_\kappa(\mathcal{D}_r(\mathbf{s}_t)))\,, \tag{3}$$

where:

- $r \in (0, 1]$ is a frequency-resolution factor,
- $\mathcal{D}_r : \mathbb{R}_+^N \to \mathbb{R}_+^{N_r}$ is a deterministic downsampling operator with

$$N_r \triangleq \lfloor rN \rfloor, \tag{4}$$

- $\Psi_\kappa : \mathbb{R}_+^{N_r} \to \mathbb{R}^{N_r}$ is an elementwise dynamic-range map,
- $\mathcal{S}_{\boldsymbol{\tau}_t} : \mathbb{R}^{N_r} \to \mathbb{R}^{N_r}$ is a blockwise standardization operator parameterized by frame-dependent side information $\boldsymbol{\tau}_t$,
- ϕ_p collects the fixed preprocessing hyperparameters.

Admissible preprocessing operators. The operators $\mathcal{D}_r$ and $\mathcal{U}_r$ are assumed to be fixed and deterministic, with

$$\mathcal{D}_r : \mathbb{R}_+^N \to \mathbb{R}_+^{N_r}, \qquad \mathcal{U}_r : \mathbb{R}_+^{N_r} \to \mathbb{R}_+^N, \tag{5}$$

and they must satisfy:

1. *nonnegativity preservation:*

$$\mathcal{D}_r(\mathbb{R}_+^N) \subseteq \mathbb{R}_+^{N_r}, \qquad \mathcal{U}_r(\mathbb{R}_+^{N_r}) \subseteq \mathbb{R}_+^N; \tag{6}$$

2. *constant-spectrum consistency:* for every $c \geq 0$,

$$\mathcal{U}_r\big(\mathcal{D}_r(c\mathbf{1}_N)\big) = c\mathbf{1}_N, \tag{7}$$

where $\mathbf{1}_N$ denotes the all-ones vector in $\mathbb{R}^N$;

3. *boundedness:* there exist finite constants $C_D, C_U > 0$ such that

$$\|\mathcal{D}_r(\mathbf{x})\|_2 \leq C_D\|\mathbf{x}\|_2, \qquad \|\mathcal{U}_r(\mathbf{y})\|_2 \leq C_U\|\mathbf{y}\|_2 \tag{8}$$

for all $\mathbf{x} \in \mathbb{R}_+^N$ and $\mathbf{y} \in \mathbb{R}_+^{N_r}$.

A concrete admissible default is local block averaging for $\mathcal{D}_r$ and linear interpolation for $\mathcal{U}_r$, but the framework itself only requires the properties above.

Dynamic-range mapping. Let

$$\mathbf{x}_t \triangleq \Psi_\kappa(\mathcal{D}_r(\mathbf{s}_t)), \qquad [\Psi_\kappa(\mathbf{v})]_n = \log(v_n + \kappa), \qquad \kappa > 0. \tag{9}$$

Its inverse is applied elementwise as

$$[\Psi_\kappa^{-1}(\mathbf{x})]_n = e^{x_n} - \kappa. \tag{10}$$

This fixes the nonlinearity explicitly while preserving invertibility.

Blockwise standardization. Partition the reduced grid into B contiguous blocks $\{\mathcal{B}_b\}_{b=1}^B$, and let $b(n)$ denote the block containing bin n. For each block, define

$$\mu_{t,b} \triangleq \frac{1}{|\mathcal{B}_b|}\sum_{n\in\mathcal{B}_b} x_{t,n}, \tag{11}$$

$$\sigma_{t,b} \triangleq \left(\frac{1}{|\mathcal{B}_b|}\sum_{n\in\mathcal{B}_b}(x_{t,n} - \mu_{t,b})^2 + \varepsilon_\sigma\right)^{1/2}, \qquad \varepsilon_\sigma > 0. \tag{12}$$

The standardized representation is

$$u_{t,n} = \frac{x_{t,n} - \mu_{t,b(n)}}{\sigma_{t,b(n)}}, \qquad \mathbf{u}_t \triangleq \begin{bmatrix} u_{t,1} & \cdots & u_{t,N_r} \end{bmatrix}^\top. \tag{13}$$

The side-information vector is

$$\boldsymbol{\tau}_t \triangleq \begin{bmatrix} \mu_{t,1} & \log\sigma_{t,1} & \cdots & \mu_{t,B} & \log\sigma_{t,B} \end{bmatrix}^\top. \tag{14}$$

We quantize $\boldsymbol{\tau}_t$ to $\hat{\boldsymbol{\tau}}_t$ using fixed scalar quantizers. If q_μ bits are used for each $\mu_{t,b}$ and q_σ bits for each $\log\sigma_{t,b}$, then the side-information rate is

$$L_\tau^{(t)} \triangleq B(q_\mu + q_\sigma). \tag{15}$$

This cost is charged explicitly and is not treated as negligible overhead.

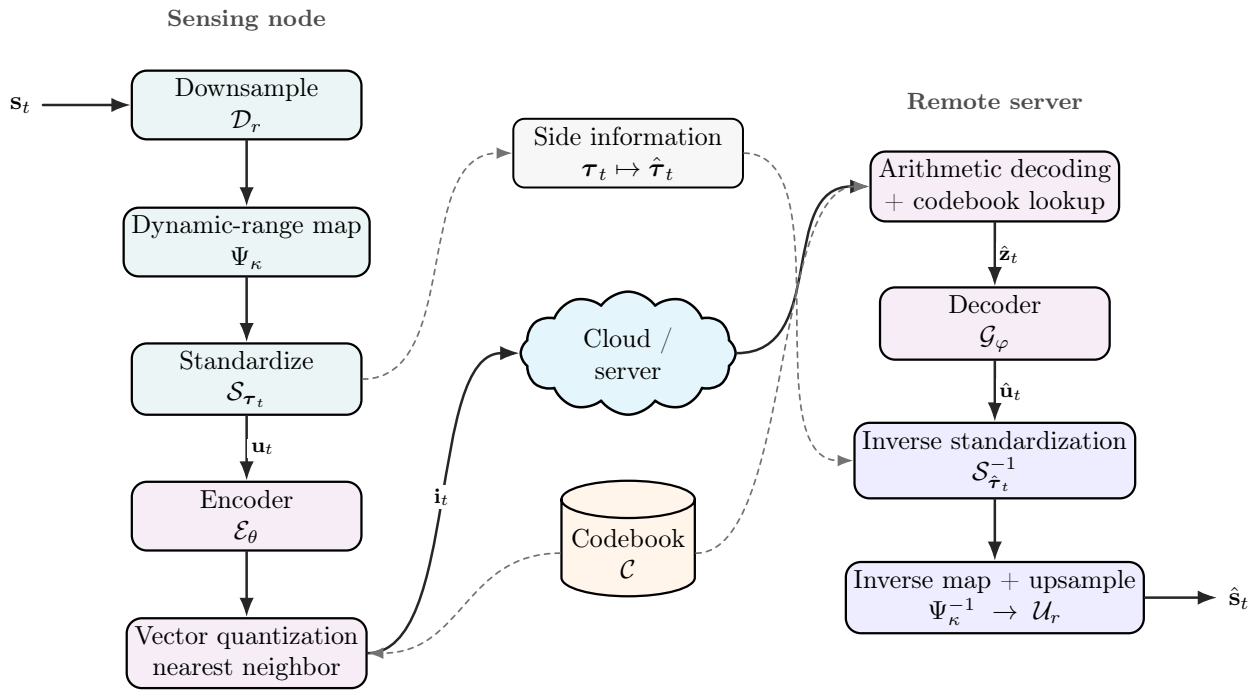

Figure 1: End-to-end workflow of the proposed PSD codec. A PSD frame $\mathbf{s}_t$ is deterministically preprocessed, encoded into discrete indices, transmitted to the cloud/server, and reconstructed as $\hat{\mathbf{s}}_t$. The side-information path $\boldsymbol{\tau}_t \mapsto \hat{\boldsymbol{\tau}}_t$ is explicit because its payload contributes to the bitrate budget through $L_\tau^{(t)}$.

Inverse preprocessing. The inverse standardization operator $\mathcal{S}^{-1}_{\hat{\boldsymbol{\tau}}_t} : \mathbb{R}^{N_r} \rightarrow \mathbb{R}^{N_r}$ is defined componentwise by

$$\left[\mathcal{S}^{-1}_{\hat{\boldsymbol{\tau}}_t}(\mathbf{u})\right]_n = \hat{\sigma}_{t,b(n)} u_n + \hat{\mu}_{t,b(n)}. \tag{16}$$

Accordingly, the inverse preprocessing chain is

$$\mathcal{T}^{-1}_{\phi_p}(\mathbf{u}; \hat{\boldsymbol{\tau}}_t) = \mathcal{U}_r\left(\Psi_\kappa^{-1}\left(\mathcal{S}^{-1}_{\hat{\boldsymbol{\tau}}_t}(\mathbf{u})\right)\right). \tag{17}$$

1.3 Learned discrete latent codec

The learned codec acts on the preprocessed frame $\mathbf{u}_t$. Let

$$\mathbf{z}_t = \mathcal{E}_\theta(\mathbf{u}_t) = \begin{bmatrix}\mathbf{z}_{t,1} & \cdots & \mathbf{z}_{t,M}\end{bmatrix}, \qquad \mathbf{z}_{t,m} \in \mathbb{R}^d, \tag{18}$$

where M is the number of latent positions and d is the latent embedding dimension.

Let the codebook be

$$\mathcal{C} = \{\mathbf{c}_1, \ldots, \mathbf{c}_J\}, \qquad \mathbf{c}_j \in \mathbb{R}^d. \tag{19}$$

Each latent vector is quantized to its nearest codeword:

$$i_{t,m} \triangleq \arg\min_{1 \le j \le J} \|\mathbf{z}_{t,m} - \mathbf{c}_j\|_2^2, \tag{20}$$

$$\hat{\mathbf{z}}_{t,m} \triangleq \mathbf{c}_{i_{t,m}}. \tag{21}$$

The discrete index sequence is

$$\mathbf{i}_t \triangleq \begin{bmatrix} i_{t,1} & \cdots & i_{t,M} \end{bmatrix}^\top, \tag{22}$$

and the corresponding quantized latent representation is

$$\hat{\mathbf{z}}_t \triangleq \begin{bmatrix} \hat{\mathbf{z}}_{t,1} & \cdots & \hat{\mathbf{z}}_{t,M} \end{bmatrix}. \tag{23}$$

The decoder reconstructs the normalized representation as

$$\hat{\mathbf{u}}_t = \mathcal{G}_\varphi(\hat{\mathbf{z}}_t), \tag{24}$$

and the final PSD reconstruction is

$$\hat{\mathbf{s}}_t = \mathcal{T}_{\phi_p}^{-1}(\hat{\mathbf{u}}_t; \hat{\boldsymbol{\tau}}_t). \tag{25}$$

During training, the non-differentiable nearest-neighbor assignment is handled with the straight-through estimator, and the vector-quantization stabilization loss is

$$\mathcal{L}_{\mathrm{VQ}}^{(t)} = \sum_{m=1}^{M} \left(\left\| \operatorname{sg}[\mathbf{z}_{t,m}] - \hat{\mathbf{z}}_{t,m} \right\|_2^2 + \beta_{\mathrm{com}} \left\| \mathbf{z}_{t,m} - \operatorname{sg}[\hat{\mathbf{z}}_{t,m}] \right\|_2^2 \right), \tag{26}$$

where $\operatorname{sg}[\cdot]$ denotes the stop-gradient operator and $\beta_{\mathrm{com}} > 0$ is the commitment weight.

1.4 Operational rate model

Because the system is a codec rather than merely an autoencoder, the rate term must be modeled explicitly. Let

$$q_\xi(\mathbf{i}_t) = \prod_{m=1}^{M} q_\xi(i_{t,m} \mid \mathbf{h}_{t,m}), \tag{27}$$

where q_ξ is an entropy model and $\mathbf{h}_{t,m}$ denotes optional causal context (e.g., previously decoded indices or side information derived from a hyperprior). A factorized model is recovered by taking $\mathbf{h}_{t,m} = \varnothing$.

The differentiable training-time rate proxy for the main index stream is

$$R_{\mathrm{idx}}^{(t)} \triangleq -\sum_{m=1}^{M} \log_2 q_\xi(i_{t,m} \mid \mathbf{h}_{t,m}). \tag{28}$$

Including side information, the total training-time rate surrogate is

$$R_{\mathrm{e}}^{(t)} \triangleq R_{\mathrm{idx}}^{(t)} + L_\tau^{(t)}. \tag{29}$$

At deployment, the sequence $\mathbf{i}_t$ is arithmetic-coded under the same model, yielding an actual coded length $L_{\mathrm{idx,ac}}^{(t)}$. The operational payload is therefore

$$L_{\mathrm{op}}^{(t)} \triangleq L_{\mathrm{idx,ac}}^{(t)} + L_\tau^{(t)}. \tag{30}$$

Reported rate–distortion curves should be based on $L_{\mathrm{op}}^{(t)}$, not merely on $R_{\mathrm{e}}^{(t)}$.

1.5 Reference noise floor and occupancy representation

Occupancy-related quantities depend on a noise-floor baseline, so that baseline must be defined externally to the codec. Let

$$\bar{\mathbf{n}}_t \triangleq \begin{bmatrix} \bar{n}_{t,1} & \cdots & \bar{n}_{t,N} \end{bmatrix}^\top \in \mathbb{R}_+^N \tag{31}$$

be a reference noise-floor estimate obtained from uncompressed frames over a sliding window of length W, with

$$\bar{n}_{t,n} \triangleq \mathcal{N}\big(\{s_{\tau,n}\}_{\tau=t-W+1}^{t}\big)\,, \tag{32}$$

where $\mathcal{N}$ is a robust baseline estimator (e.g., a lower percentile, trimmed mean, or calibration-based rule).

To avoid conflating compression error with baseline-estimation error, the same $\bar{\mathbf{n}}_t$ and threshold margin γ are used to define occupancy from both $\mathbf{s}_t$ and $\hat{\mathbf{s}}_t$ during training and primary evaluation. For differentiability, define

$$\sigma_{\tau_{\text{occ}}}(x) = \frac{1}{1+\exp(-x/\tau_{\text{occ}})}, \qquad \tau_{\text{occ}} > 0, \tag{33}$$

and the soft occupancies

$$p_{t,n} \triangleq \sigma_{\tau_{\text{occ}}}(s_{t,n} - \bar{n}_{t,n} - \gamma), \tag{34}$$

$$\hat{p}_{t,n} \triangleq \sigma_{\tau_{\text{occ}}}(\hat{s}_{t,n} - \bar{n}_{t,n} - \gamma). \tag{35}$$

Their hard-thresholded versions are

$$o_{t,n} = \mathbf{1}\{p_{t,n} \geq 1/2\}, \qquad \hat{o}_{t,n} = \mathbf{1}\{\hat{p}_{t,n} \geq 1/2\}. \tag{36}$$

1.6 Core distortion and optional task regularization

The framework separates a core codec objective from an optional task regularizer.

Core PSD fidelity. The core distortion term is

$$D_{\text{psd}}^{(t)} = \frac{1}{N}\sum_{n=1}^{N}\left[\log(s_{t,n}+\kappa) - \log(\hat{s}_{t,n}+\kappa)\right]^2. \tag{37}$$

This term is task-agnostic and measures reconstruction fidelity in the spectral domain.

Generic task map. Let

$$\phi_{\text{task}} : \mathbb{R}_+^N \times \mathbb{R}_+^N \to \mathbb{R}^q \tag{38}$$

be a generic task map, and define

$$\mathbf{f}_t = \phi_{\text{task}}(\mathbf{s}_t; \bar{\mathbf{n}}_t), \qquad \hat{\mathbf{f}}_t = \phi_{\text{task}}(\hat{\mathbf{s}}_t; \bar{\mathbf{n}}_t). \tag{39}$$

The corresponding abstract task regularizer is denoted by

$$D_{\text{task}}^{(t)} \triangleq \mathfrak{D}_{\text{task}}(\mathbf{s}_t, \hat{\mathbf{s}}_t; \bar{\mathbf{n}}_t), \tag{40}$$

and may represent occupancy consistency, feature preservation, a downstream task head, or be omitted entirely.

Illustrative sensing instantiation (example only). One admissible instantiation is a combination of occupancy consistency and feature preservation. This block is illustrative only and is not a privileged default of the framework.

First, define the occupancy consistency term

$$D_{\mathrm{occ}}^{(t)} = -\frac{1}{N}\sum_{n=1}^{N}\left[w_1 p_{t,n}\log\hat{p}_{t,n} + w_0(1-p_{t,n})\log(1-\hat{p}_{t,n})\right], \tag{41}$$

where $w_1, w_0 > 0$ allow asymmetric penalties for missed detections and false alarms.

Second, let

$$\mathcal{H} : \mathbb{R}_+^N \to \mathbb{R}_+^N \tag{42}$$

be a fixed deterministic smoothing operator used only for feature extraction. We assume that $\mathcal{H}$ preserves nonnegativity and constant spectra, i.e.,

$$\mathcal{H}(\mathbb{R}_+^N) \subseteq \mathbb{R}_+^N, \qquad \mathcal{H}(c\mathbf{1}_N) = c\mathbf{1}_N \quad \text{for all } c \geq 0. \tag{43}$$

A concrete admissible default is a normalized moving-average filter with odd window length $2\ell+1$.

Define the illustrative feature vector

$$\mathbf{f}_t^{\mathrm{ex}} \triangleq \begin{bmatrix} f_t^{\mathrm{pk}} & f_t^{\mathrm{cent}} & f_t^{\mathrm{bw}} \end{bmatrix}^\top, \tag{44}$$

where:

- f_t^{pk} is the dominant spectral peak location computed after smoothing,

$$\tilde{\mathbf{s}}_t \triangleq \mathcal{H}(\mathbf{s}_t), \qquad n_t^\star \triangleq \operatorname*{arg\,max}_{1\leq n\leq N} \tilde{s}_{t,n}, \qquad f_t^{\mathrm{pk}} \triangleq \omega_{n_t^\star}; \tag{45}$$

- f_t^{cent} is the spectral centroid,

$$f_t^{\mathrm{cent}} \triangleq \frac{\sum_{n=1}^N \omega_n s_{t,n}}{\sum_{n=1}^N s_{t,n}}; \tag{46}$$

- f_t^{bw} is the occupied bandwidth of a selected connected occupied component.

Let

$$\mathfrak{C}_t \triangleq \{\mathcal{I} \subseteq \Omega : \mathcal{I} \text{ is a connected component of } \{\omega_n : o_{t,n} = 1\}\}. \tag{47}$$

If $\mathfrak{C}_t \neq \varnothing$, define the selected component by

$$\mathcal{I}_t \triangleq \operatorname*{arg\,max}_{\mathcal{I}\in\mathfrak{C}_t} \sum_{\{n:\,\omega_n\in\mathcal{I}\}} s_{t,n}, \tag{48}$$

that is, the occupied connected component with largest integrated spectral mass. If $\mathfrak{C}_t = \varnothing$, we set

$$f_t^{\mathrm{bw}} \triangleq 0. \tag{49}$$

Otherwise,

$$f_t^{\mathrm{bw}} \triangleq \max_{\omega_n\in\mathcal{I}_t} \omega_n - \min_{\omega_n\in\mathcal{I}_t} \omega_n. \tag{50}$$

Define $\hat{\mathbf{f}}_t^{\mathrm{ex}}$ analogously from $\hat{\mathbf{s}}_t$, using the same operator $\mathcal{H}$ and the analogous component-selection rule on the reconstructed occupied set $\{\omega_n : \hat{o}_{t,n} = 1\}$.

Using the Huber penalty

$$\rho_\delta(x) = \begin{cases} \frac{1}{2}x^2, & |x| \le \delta, \\ \delta\left(|x| - \frac{\delta}{2}\right), & |x| > \delta, \end{cases} \tag{51}$$

define the feature-preservation term

$$D_{\text{feat}}^{(t)} = \alpha_{\text{pk}}\,\rho_\delta(f_t^{\text{pk}} - \hat{f}_t^{\text{pk}}) + \alpha_{\text{cent}}\,\rho_\delta(f_t^{\text{cent}} - \hat{f}_t^{\text{cent}}) + \alpha_{\text{bw}}\,\rho_\delta(f_t^{\text{bw}} - \hat{f}_t^{\text{bw}}). \tag{52}$$

The resulting illustrative task regularizer is

$$D_{\text{task,ex}}^{(t)} \triangleq \beta_{\text{occ}} D_{\text{occ}}^{(t)} + \beta_{\text{feat}} D_{\text{feat}}^{(t)}. \tag{53}$$

1.7 Rate–distortion learning problem

Let

$$\Theta \triangleq (\theta, \varphi, \mathcal{C}, \xi) \tag{54}$$

collect the learned encoder, decoder, codebook, and entropy-model parameters. The operational design problem is

$$\min_{\Theta} \quad \mathbb{E}\Big[\beta_{\text{psd}} D_{\text{psd}}^{(t)} + \beta_{\text{task}} D_{\text{task}}^{(t)}\Big] \qquad \text{subject to} \qquad \mathbb{E}\Big[L_{\text{op}}^{(t)}\Big] \le R_{\max}, \tag{55}$$

where $R_{\max}$ is the target average payload per frame. The task-agnostic case is recovered by setting $\beta_{\text{task}} = 0$. If the illustrative sensing instantiation is adopted, one simply sets

$$D_{\text{task}}^{(t)} = D_{\text{task,ex}}^{(t)}. \tag{56}$$

In practice, training uses the Lagrangian surrogate

$$\mathcal{L}^{(t)} = \beta_{\text{psd}} D_{\text{psd}}^{(t)} + \lambda_R R_{\text{e}}^{(t)} + \eta \mathcal{L}_{\text{VQ}}^{(t)} + \beta_{\text{task}} D_{\text{task}}^{(t)}, \tag{57}$$

where $\lambda_R > 0$ controls the rate–distortion trade-off and $\eta > 0$ weights the vector-quantization stabilization term.

1.8 Preprocessing-only reference for ablation

Because deterministic preprocessing can itself introduce distortion, the framework explicitly separates preprocessing loss from learned compression loss. Define the preprocessing-only reconstruction by

$$\bar{\mathbf{s}}_t \triangleq \mathcal{T}_{\phi_p}^{-1}\big(\mathcal{T}_{\phi_p}(\mathbf{s}_t; \boldsymbol{\tau}_t); \hat{\boldsymbol{\tau}}_t\big)\,. \tag{58}$$

Then, for any chosen evaluation distortion

$$\mathscr{D} : \mathbb{R}_+^N \times \mathbb{R}_+^N \to \mathbb{R}_+, \tag{59}$$

define

$$\Delta_{\text{pre}}^{(t)} \triangleq \mathscr{D}(\mathbf{s}_t, \bar{\mathbf{s}}_t), \tag{60}$$

$$\Delta_{\text{codec}}^{(t)} \triangleq \mathscr{D}(\bar{\mathbf{s}}_t, \hat{\mathbf{s}}_t). \tag{61}$$

The term $\Delta_{\text{pre}}^{(t)}$ captures distortion introduced solely by deterministic preprocessing and side-information quantization, whereas $\Delta_{\text{codec}}^{(t)}$ captures the incremental distortion introduced by the learned discrete codec. This separation is essential for ablation and for interpreting whether the learned bottleneck is actually carrying the compression burden.